

Dominique C. Perrault-Joncas

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Summary

Principal Scientist with 13+ years applying reinforcement learning, causal inference and evaluation methodology, and LLM post-training (RLHF / preference optimization) to large-scale production decision systems. Led tech for cross-functional teams (15+), published at top venues (NeurIPS, TMLR), and delivered multiple nine-figure impact programs across experimentation, inventory placement, and forecasting at Amazon and Google.

Selected Impact

- Led tech for RL initiatives driving investment in engineering teams to scale a **nine-figure** opportunity in North America automated ordering systems.
- Demonstrated capacity-constrained RL ordering improving profit **+1.8%** under **10%** tighter constraints vs. production.
- Refined product launch criteria via meta-analysis of randomized experiments; backtests indicate **nine-figure** incremental impact vs. prior rules.
- Delivered an experimentation framework that reduced the risk of failures by $10\times$, mitigating potential **eight-figure** downside per failed test.

Technical Skills

Reinforcement Learning: Model-based RL, constrained RL, sim2real

LLM Post-Training: RLHF, preference optimization (DPO / C2-DPO), fine-tuning, evaluation

Game Theory & Market Design: marketplace equilibrium, competitive pricing mechanisms

Causal & Experimentation: interference-aware experimentation under constraints, optimal treatment effects estimation, Heavy-tailed Outcome Analysis

Engineering: Python, PyTorch, SQL/Spark, AWS

Experience

Amazon Jan 2020 – Present | Seattle, WA

Principal Research Scientist, Supply Chain Optimization Technologies

- **RL for Inventory Buying & Placement (Tech Lead, 2024–Present):** Co-led a team of 15+ scientists and engineers building reinforcement learning systems for inventory buying and placement decisions spanning multi-billion-dollar annual inventory spend; success drove follow-on investment in dedicated engineering teams to scale a nine-figure opportunity in North America.
- **Capacity-Constrained RL Buying:** Developed RL buying methods under capacity constraints; demonstrated +1.8% profit improvement even with 10% tighter buying constraints versus production baselines.
- **RL Architecture & Sim2Real:** Defined action space and value function design for RL-based placement, including a quadratic coordination interface across subsystems; launched first production labs for placement RL agents. Directed sim2real evaluation frameworks bridging simulation-to-production gaps.
- **Experimentation under Joint Constraints:** Developed and deployed an interference-aware experimentation methodology for resource-constrained environments; reduced risk of unsuccessful scaled experiments by $10\times$, mitigating potential eight-figure downside per failed large-scale test.
- **Meta-analysis & Launch Decisions:** Led meta-analysis of randomized experiments to improve treatment effect estimation for heavy-tailed outcomes; backtests across multiple orgs indicate updated launch decision criteria would have delivered nine-figure incremental impact versus prior rules (NeurIPS 2022 Workshop, with M.I. Jordan).

- **LLM Post-training / Preference Optimization:** Co-developed a unifying “implicit classification” framework for DPO-style algorithms; identified under-specification leading to winner–loser probability collapse and proposed C2-DPO, a constrained preference optimization method with an RLHF interpretation that improves alignment quality on standard preference datasets and multiple LLM checkpoints (arXiv 2025).
- **LLM-Augmented Forecasting:** Co-developed LLMForecaster, integrating unstructured text into seasonal demand forecasting via LLM embeddings (NeurIPS 2024 Workshop).
- **Leadership:** Tech lead for 15+ researcher RL group; built advisory committee governing experimental platform changes; Senior Area Chair for Amazon Consumer Science Summit (2021–26).

Google

Apr 2015 – Dec 2019 | Seattle, WA

Senior Quantitative Analyst / Statistician, Advanced Measurement Technologies

- **Cross-Media Measurement (Tech Lead, 2017–2019):** Led a team of 3 statisticians on a cross-media measurement panel initiative. Designed experiments for face recognition and gaze estimation systems for production deployment.
- **Indoor Positioning & Prediction:** Developed indoor positioning models using WiFi, Bluetooth, and sensor fusion. Built box office prediction models; designed methods for imputing missing user data at scale.

Amazon

Nov 2012 – Apr 2015 | Seattle, WA

Research Scientist, Demand Forecasting — built demand forecasting from the ground up for the Amazon Pantry launch; quantile regression and forecasting under censored demand.

Submitted / Under Review

- H. Parikh, . . . , D. Madeka, M.I. Jordan, D. Foster, **D. Perrault-Joncas**, A. Volfovsky. *Towards Optimal Estimators for Randomized Control Trials*. Submitted to **JASA**; [arXiv](#), 2026.
- H. Parikh, G. Levin-Konigsberg, **D. Perrault-Joncas**, A. Volfovsky. *Mind the Sim-to-Real Gap & Think Like a Scientist*. Submitted to **NeurIPS 2026**; [arXiv](#).
- A. Wang, **D. Perrault-Joncas**, . . . , D. Foster, C. Eisenach. *Ready from Day 1: Population-Aware Coordination for Large-Scale Constrained Multi-Agent Systems*. Submitted to **NeurIPS 2026**; [arXiv](#).
- H. Waldon, C. Eisenach, A. Bagaria, D. Russo, **D. Perrault-Joncas**, . . . , D. Foster. *Proximal Residual Value Functions for Consistent Planning and Real-Time Execution*. Submitted to **NeurIPS 2026 Workshop (ML×OR)**; [arXiv](#).

Selected Publications

- D. Holtz, I. Bojinov, R. Johari, N. Kallus, . . . , **D. Perrault-Joncas**, . . . , M. Tingley. *The Future of Large-Scale Experiments and Their Challenges in the Digital Era*. **Nature Human Behaviour**, 2026.
- R.J. George, C. Eisenach, U. Ghai, **D. Perrault-Joncas**, . . . , D. Foster. *BRIDGE: Building Representations In Domain Guided Program Synthesis*. **ICML 2026 Workshop (AI4Math)**; [OpenReview](#).
- K. Asadi, X. Xu, . . . , **D. Perrault-Joncas**, . . . , M. Ghavamzadeh. *C2-DPO: Constrained Controlled Classification DPO*. Accepted at **TMLR**, 2025. [OpenReview](#).
- A. Maggiar, . . . , **D. Perrault-Joncas**. *Structure-Informed Deep RL for Inventory Management*. **NeurIPS 2025 Workshop**.
- H. Zhang, . . . , **D. Perrault-Joncas**, . . . , A.G. Wilson. *LLMForecaster: Improving Seasonal Event Forecasts with Unstructured Textual Data*. **NeurIPS 2024 Workshop**.
- N. Tripuraneni, **D. Perrault-Joncas**, D. Madeka, D. Foster, M.I. Jordan. *Meta-Analysis of Randomized Experiments with Applications to Heavy-Tailed Response Data*. **NeurIPS 2022 Workshop**.
- T. Ding, **D. Perrault-Joncas**, O. Ronen, M.I. Jordan, D. Bergemann, D. Foster, O. Gottesman. *Marketplace Operators Can Induce Competitive Pricing*. [arXiv](#), 2025.

Also: 3 NeurIPS manifold-learning papers (2011–2017); 25+ internal Amazon papers.

Education: Ph.D. Statistics, UW (M. Meilă), 2012 | M.Sc. Applied Math, McGill, 2007